The one-phase Muskat flow and the Hele–Shaw semi-flow

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Abstract. In this paper, we identify the evolution constructed in [2] with the Hele–Shaw semi-flow introduced in [1], and establish eventual analyticity together with expontential decay of the Lipschitz seminorm.

1 Introduction and main results

The one-phase Muskat problem describes the motion of a fluid in a porous medium governed by Darcy's law. After a suitable normalization, Darcy's law takes the form

$$u = -\nabla P - e_N, \qquad \operatorname{div} u = 0 \quad \text{in } \mathcal{O}_t, \tag{1.1}$$

where u is the fluid velocity, P is the fluid pressure, $\mathcal{O}_t$ is the fluid domain, $e_N = (0, \dots, 0, 1)$, and $-e_N$ represents the gravitational force. We consider the zero surface-tension regime:

$$P = 0 \quad \text{on } \Sigma_t. \tag{1.2}$$

The motion of the free boundary $\Sigma_t = \partial\mathcal{O}_t$ is determined by the kinematic condition

$$V(\Sigma_t) = u \cdot n, \tag{1.3}$$

where $V(\Sigma_t)$ denotes the normal velocity of Σ_t and n is the outward-pointing unit normal to Σ_t. When the fluid domain is the subgraph of a function $h(t, x)$, the evolution can be written as (see § 2.4),

$$\partial_t h = -G(h)h. \tag{1.4}$$

Dong, Gancedo, and Nguyen [2] constructed a global Lipschitz solution to (1.4) in the strong $L^\infty_t L^2_x$ sense and showed that it is the unique viscosity solution. Nguyen [3] later established exponential decay of its C^α norm as $t \to \infty$ for every $0 < \alpha < 1$.

A different description of the evolution was developed in [1] through an obstacle formulation and a corresponding semi-flow. For smooth solutions, [1] showed that this semi-flow agrees with the free boundary evolution (1.4). Our first aim is to establish the same identification in the Lipschitz setting. Once this identification is established, the regularity theory in [1], together with the decay estimate in [3], shows that the solution constructed in [2] is eventually analytic and that its Lipschitz norm decays. The corresponding identification in $\mathbb{R}^N$, $N \geq 3$, or for rougher initial data is not known (see [4]). Our main results are as follows.

We use the same notation for periodic functions and sets on $\mathbb{T} \times \mathbb{R}$ and their periodic lifts to $\mathbb{R}^2$.

Main Theorem. *Let $h_0 \in W^{1,\infty}(\mathbb{T})$, let h be the global Lipschitz solution given by Proposition 2.1, and let Φ be the semi-flow given by Proposition 2.2. Let $\mathcal{O}_t := \{(x, y) \in \mathbb{T} \times \mathbb{R} : y < h(t, x)\}$. Then for every $t \geq 0$,*

$$\mathcal{O}_t = \Phi(t, \mathcal{O}_0). \tag{1.5}$$

Moreover, $h(t, \cdot)$ is real analytic for every

$$t > \operatorname{osc} h_0 := \sup h_0 - \inf h_0, \tag{1.6}$$

and there exist $T, \lambda, C > 0$, depending on the initial data, such that

$$\|\partial_x h(t)\|_{L^\infty(\mathbb{T})} \leq Ce^{-\lambda t}, \qquad t \geq T. \tag{1.7}$$

2 Preliminaries

In this section, we collect the ingredients needed for the proof of the main theorem. We first recall the relevant results from [2] and [1] in § 2.1 and § 2.2, respectively. In § 2.3, we introduce the operator appearing in (1.4). Finally, in § 2.4 and § 2.5, we review the tools from [1] used to formulate the obstacle problem.

2.1 Global Lipschitz solutions

Proposition 2.1. *For every $h_0 \in W^{1,\infty}(\mathbb{T})$, there exists*

$$h \in C(\mathbb{T}\times[0,\infty)) \cap L^\infty([0,\infty);W^{1,\infty}(\mathbb{T})), \qquad \partial_t h \in L^\infty([0,\infty);L^2(\mathbb{T})), \tag{2.1}$$

such that $h|_{t=0} = h_0$. Moreover, h satisfies (1.4) in $L^\infty_t L^2_x$, and

$$\|h(t)\|_{L^\infty(\mathbb{T})} \le \|h_0\|_{L^\infty(\mathbb{T})}, \qquad \|\partial_x h(t)\|_{L^\infty(\mathbb{T})} \le \|\partial_x h_0\|_{L^\infty(\mathbb{T})}. \tag{2.2}$$

2.2 The semi-flow

We write $H^1_{\mathrm{loc},+}(\mathbb{R}^N) = \{u \in H^1_{\mathrm{loc}}(\mathbb{R}^N) : u \ge 0\}$.

Proposition 2.2. *Let $N \ge 2$ and set $d = N-1$.*

1. *Consider an open connected set $\mathcal{O}_0 \subset \mathbb{R}^N$ such that, for some constant $C > 0$,*
$$\{x \in \mathbb{R}^N : x_N < -C\} \subset \mathcal{O}_0 \subset \{x \in \mathbb{R}^N : x_N < C\}. \tag{2.3}$$
Then, for every $t > 0$, there exists a unique variational solution $v = v(t) \in H^1_{\mathrm{loc},+}(\mathbb{R}^N)$ to the obstacle problem
$$\Delta v(t) = \mathbf{1}_{\{v(t)>0\}\cap(\mathbb{R}^N\setminus\mathcal{O}_0)}, \tag{2.4}$$
satisfying
$$v_0(t, x_N + C) \le v(t,x) \le v_0(t, x_N - C) \qquad \text{for every } x \in \mathbb{R}^N,$$
where
$$v_0(t, x_N) := \int_0^t \max\{0, -x_N + \tau\}\, d\tau.$$
2. *The set $\{v(t) > 0\}$ is connected for every $t > 0$.*
3. *We denote*
$$\mathcal{U} = \big\{\Omega \subset \mathbb{R}^N : \Omega \text{ is open and connected and satisfies (2.3) for some } C > 0\big\}.$$
For $t > 0$, set
$$\Phi(t, \mathcal{O}_0) := \{v(t) > 0\} - te_N,$$
and set $\Phi(0, \mathcal{O}_0) := \mathcal{O}_0$. Then $\Phi : [0,\infty)\times\mathcal{U} \to \mathcal{U}$ is a semi-flow, i.e.
$$\Phi\big(t, \Phi(s, \mathcal{O}_0)\big) = \Phi(t+s, \mathcal{O}_0) \quad \text{for } s, t \ge 0,\ \mathcal{O}_0 \in \mathcal{U}.$$

2.3 Dirichlet–Neumann operator

For a function $h : [0,\infty)\times\mathbb{T}^d \to \mathbb{R}$, we still write $\mathcal{O}_t = \{(x,y) \in \mathbb{T}^d \times \mathbb{R} : y < h(t,x)\}$.

In this subsection, we suppress t and write $\mathcal{O} = \mathcal{O}_t$ for simplicity.

Given a function $g : \mathbb{T}^d \to \mathbb{R}$, $d \geq 1$, we consider the boundary value problem

$$\begin{cases} \Delta_{x,y}\phi = 0 \quad \text{in } \mathcal{O}, \\ \phi(x, h(x)) = g(x), \quad \nabla_{x,y}\phi \in L^2(\mathcal{O}). \end{cases} \tag{2.5}$$

The Dirichlet–Neumann operator $G(h)$ is defined by

$$G(h)g = (-\nabla_x h, 1) \cdot \nabla_{x,y}\phi|_{y=h(x)}. \tag{2.6}$$

In general, $G(h)g$ need not be classically defined, but it can be viewed as an element of $H^{-1/2}(\mathbb{T}^d)$.

Proposition 2.3. *Let $h \in \mathrm{Lip}(\mathbb{T}^d)$. For any $g \in \dot{H}^{1/2}(\mathbb{T}^d)$, there exists a unique solution $\phi \in \dot{H}^1(\mathcal{O})$ to (2.5), where*

$$\dot{H}^1(\mathcal{O}) = \{u \in L^1_{\mathrm{loc}}(\mathcal{O}) : \nabla_{x,y}u \in L^2(\mathcal{O})\}/\mathbb{R}$$

is endowed with the norm $\|u\|_{\dot{H}^1(\mathcal{O})} = \|\nabla u\|_{L^2(\mathcal{O})}$. Moreover, we have $G(h)g \in H^{-1/2}(\mathbb{T}^d)$, given by

$$\langle G(h)g, \xi \rangle_{H^{-1/2}(\mathbb{T}^d), H^{1/2}(\mathbb{T}^d)} = \int_{\mathcal{O}} \nabla\phi \cdot \nabla E\xi, \tag{2.7}$$

where $\xi \in H^{1/2}(\mathbb{T}^d)$ and $E\xi \in \dot{H}^1(\mathcal{O})$ is any extension of ξ with $\operatorname{Tr} E\xi = \xi$.

We defer the proof to Appendix A.

2.4 Free-falling coordinates

We introduce free-falling coordinates to obtain fluid domains that are nondecreasing in time. Under the change of variables $z = y + t$, the gravitational term disappears from Darcy's law, and the transformed velocity satisfies $\widetilde{u} = u + e_N = -\nabla p$. By the maximum principle, the pressure is positive in the fluid region. Since it vanishes on the free boundary, we formally expect the boundary to move outward in these coordinates.

More specifically, by (1.1) and (1.2), setting $\phi(t, x, y) = P(t, x, y) + y$, we have

$$\begin{cases} \Delta_{x,y}\phi = 0 \quad \text{in } \mathcal{O}_t, \\ \phi(x, h(t, x)) = h(t, x). \end{cases}$$

By the kinematic condition, we have

$$\frac{\partial_t h}{\sqrt{1 + |\nabla_x h|^2}} = u \cdot n = -\partial_n \phi.$$

Hence, by (2.6),

$$\partial_t h = -\sqrt{1 + |\nabla_x h|^2}\, \partial_n \phi = -G(h)h.$$

Passing to the free-falling coordinates $z = y + t$, set

$$\begin{aligned} &F(t, x) := h(t, x) + t, \\ &\Omega_t := \{(x, z) \in \mathbb{T}^d \times \mathbb{R} : z < F(t, x)\} = \mathcal{O}_t + te_N. \end{aligned} \tag{2.8}$$

Correspondingly, we set $p(t, x, z) = P(t, x, z - t)$. Since the Dirichlet–Neumann operator is invariant under vertical translations and annihilates constants, we have

$$\partial_t F = \partial_t h + 1 = -G(h)h + 1 = 1 - G(F)F. \tag{2.9}$$

In Section 3, we will show that $\partial_t F \geq 0$ and hence $\Omega_s \subset \Omega_t$ whenever $0 \leq s \leq t$.

2.5 Baiocchi–Duvaut transformation

To relate the evolution to the obstacle problem, we consider the time integral of the pressure. Extend $p(t)$ by zero outside Ω_t and, for a fixed $T > 0$, define

$$v(t, x, z) := \int_0^t p(s, x, z)\, ds, \qquad 0 \le t \le T. \tag{2.10}$$

This is the Baiocchi–Duvaut transformation. The joint continuity of p, established in Proposition 3.1, makes the integral well defined at every point.

The positivity set of $v(t)$ records the region occupied by the fluid up to time t. Since the domains are nondecreasing in time, we expect

$$\{v(t) > 0\} = \Omega_t, \qquad t > 0.$$

In the next section, we will prove this identity and show that $v(t)$ solves the obstacle problem (2.4). Thus the fluid domain at each time can be recovered from a static obstacle problem.

3 The obstacle problem

Let $h_0 \in W^{1,\infty}(\mathbb{T})$, and let h and $\mathcal{O}_t$ be as in Main Theorem. Choose $C > 0$ such that $\|h_0\|_{L^\infty(\mathbb{T})} \le C$. By the maximum principle, $\|h(t)\|_{L^\infty(\mathbb{T})} \le C$ for all $t \ge 0$. Clearly, $\Omega_0 = \mathcal{O}_0$ satisfies (2.3). Our goal is to show that the Baiocchi–Duvaut transformation of the associated pressure solves the obstacle problem (2.4).

Let $\mathcal{M} = \mathbb{T} \times \mathbb{R}$. We also record the joint continuity and elementary properties of the extended pressure. The proof is given in Appendix B.

Proposition 3.1. *The zero-extended pressure admits a representative, still denoted by p, such that*

$$p \in C\big([0, \infty) \times \mathcal{M}\big). \tag{3.1}$$

For every $t \ge 0$, $p(t) \in H^1_{\mathrm{loc}}(\mathcal{M})$,

$$p(t) \ge 0 \quad \text{on } \mathcal{M}, \qquad \{p(t) > 0\} = \Omega_t, \tag{3.2}$$

and

$$\Delta p(t) \ge 0 \qquad \text{in } \mathcal{D}'(\mathcal{M}). \tag{3.3}$$

Moreover, for every $t \ge 0$,

$$(t - z - C)_+ \le p(t, x, z) \le (t - z + C)_+ \qquad \text{for every } (x, z) \in \mathcal{M}. \tag{3.4}$$

To derive the obstacle problem, we first compute $\Delta p(t)$. In the smooth free-boundary case, Green's formula and the kinematic condition give

$$\begin{aligned} \langle \Delta p(t), \varphi \rangle &= \int_{\Omega_t} p(t) \Delta \varphi = -\int_{\partial\Omega_t} \varphi\, \partial_n p(t)\, dS \\ &= \int_{\partial\Omega_t} \varphi\, \frac{\partial_t F}{\sqrt{1 + |\partial_x F|^2}}\, dS = \int_{\mathbb{T}} \partial_t F(t, x)\, \varphi\big(x, F(t, x)\big)\, dx. \end{aligned}$$

In the Lipschitz setting, since $\partial_n p$ may not be defined classically, we express the corresponding boundary flux in terms of the Dirichlet–Neumann operator to complete the calculation.

Lemma 3.2. *For almost every $t > 0$ and every $\varphi \in C_c^\infty(\mathbb{T} \times \mathbb{R})$,*

$$\langle \Delta p(t), \varphi \rangle = \int_{\mathbb{T}} \partial_t F(t,x)\, \varphi\big(x, F(t,x)\big)\, dx. \tag{3.5}$$

In particular,

$$\partial_t F(t,x) \geq 0 \quad \text{for a.e. } (t,x) \in (0,\infty) \times \mathbb{T}, \tag{3.6}$$

and for every $0 \leq s \leq t$,

$$\Omega_s \subset \Omega_t.$$

Proof. Set $\bar{\varphi}(x,y) = \varphi(x, y+t)$. By Proposition 2.3 and a change of variables, we obtain

$$\begin{aligned}
\langle \Delta p(t), \varphi \rangle &= -\int_{\Omega_t} \nabla p(t) \cdot \nabla \varphi \, dx\, dz \\
&= -\int_{\mathcal{O}_t} \nabla P(t) \cdot \nabla \bar{\varphi} \, dx\, dy \\
&= -\int_{\mathcal{O}_t} \nabla \phi(t) \cdot \nabla \bar{\varphi} \, dx\, dy + \int_{\mathcal{O}_t} \nabla \bar{\varphi} \cdot e_2 \, dx\, dy \\
&= -\langle G(h)h, \bar{\varphi}|_{\partial \mathcal{O}_t} \rangle_{H^{-1/2}, H^{1/2}} + \int_{\mathbb{T}} \bar{\varphi}\big(x, h(t,x)\big)\, dx \\
&= \langle 1 - G(h)h, \bar{\varphi}|_{\partial \mathcal{O}_t} \rangle_{H^{-1/2}, H^{1/2}} \\
&= \langle \partial_t F(t), \bar{\varphi}|_{\partial \mathcal{O}_t} \rangle_{H^{-1/2}, H^{1/2}}.
\end{aligned}$$

By the regularity in (2.1), the last duality pairing is the usual integral pairing, giving

$$\langle \Delta p(t), \varphi \rangle = \int_{\mathbb{T}} \partial_t F(t,x)\, \varphi\big(x, F(t,x)\big)\, dx.$$

To obtain (3.6), choose $\varphi(x,z) = \eta(x)\chi(z)$, where $\eta \in C^\infty(\mathbb{T})$ is nonnegative and χ is a nonnegative cutoff satisfying $\chi = 1$ on a bounded interval containing the range of $F(t,\cdot)$. By the subharmonicity of $p(t)$ from the previous proposition, we have

$$\langle \Delta p(t), \varphi \rangle = \int_{\mathbb{T}} \partial_t F(t,x) \eta(x)\, dx \geq 0$$

for every nonnegative $\eta \in C^\infty(\mathbb{T})$. Hence, $\partial_t F(t,\cdot) \geq 0$ almost everywhere.

By (2.1), we have $F \in W^{1,1}_{\mathrm{loc}}((0,\infty) \times \mathbb{T})$, and thus the map $t \mapsto F(t,x)$ is absolutely continuous and nondecreasing for almost every x. By continuity of F, $F(s,x) \leq F(t,x)$ for every $x \in \mathbb{T}$ and $0 \leq s \leq t$. Therefore, $\Omega_s \subset \Omega_t$. □

Proposition 3.3 (The obstacle problem)**.** *For each $t > 0$, let*

$$v(t,x,z) = \int_0^t p(s,x,z)\, ds.$$

Then $v(t) \in H^1_{\mathrm{loc},+}(\mathcal{M})$,

$$\{v(t) > 0\} = \Omega_t, \tag{3.7}$$

and

$$\Delta v(t) = \mathbf{1}_{\{v(t)>0\} \cap (\mathcal{M} \setminus \Omega_0)} \qquad \text{in } \mathcal{D}'(\mathcal{M}). \tag{3.8}$$

Moreover,

$$v_0(t, z+C) \leq v(t,x,z) \leq v_0(t, z-C) \qquad \text{for every } (x,z) \in \mathcal{M},$$

where

$$v_0(t,z) := \int_0^t \max\{0, -z+\tau\}\, d\tau.$$

Consequently, the periodic lift of $v(t)$ to $\mathbb{R}^2$ solves the obstacle problem (2.4).

Proof. Fix $t > 0$. By the preceding results, $v(t)$ is well defined in $L^1_{\mathrm{loc}}(\mathcal{M})$. Hence, for every $\varphi \in C_c^\infty(\mathcal{M})$,

$$\begin{aligned}\langle \Delta v(t), \varphi \rangle &= \int_{\mathbb{T}} \int_{\mathbb{R}} v(t,x,z)\, \Delta\varphi(x,z)\, dz\, dx \\ &= \int_0^t \int_{\mathbb{T}} \int_{\mathbb{R}} p(s,x,z)\, \Delta\varphi(x,z)\, dz\, dx\, ds \\ &= \int_0^t \langle \Delta p(s), \varphi \rangle\, ds \\ &= \int_0^t \int_{\mathbb{T}} \partial_s F(s,x)\, \varphi(x, F(s,x))\, dx\, ds.\end{aligned}$$

By the absolute continuity of $s \mapsto F(s,x)$,

$$\begin{aligned}\langle \Delta v(t), \varphi \rangle &= \int_{\mathbb{T}} \int_{F(0,x)}^{F(t,x)} \varphi(x,z)\, dz\, dx \\ &= \int_{\mathbb{T}} \int_{\mathbb{R}} (\mathbf{1}_{\Omega_t} - \mathbf{1}_{\Omega_0}) \varphi(x,z)\, dz\, dx.\end{aligned}$$

Therefore,

$$\Delta v(t) = \mathbf{1}_{\Omega_t} - \mathbf{1}_{\Omega_0} \qquad \text{in } \mathcal{D}'(\mathcal{M}). \tag{3.9}$$

By elliptic regularity,

$$v(t) \in W^{2,q}_{\mathrm{loc}}(\mathcal{M}) \qquad \text{for every } 1 < q < \infty.$$

In particular,

$$v(t) \in H^1_{\mathrm{loc}}(\mathcal{M}), \qquad v(t) \in C^{1,\alpha}_{\mathrm{loc}}(\mathcal{M}) \quad \text{for every } 0 < \alpha < 1.$$

To prove (3.7), first let $X \notin \Omega_t$. Since $\Omega_s \subset \Omega_t$ for $0 \le s \le t$, we have $p(s,X) = 0$ throughout this interval, and hence $v(t,X) = 0$. If $X \in \Omega_t$, then $p(t,X) > 0$. By continuity, there exists $\varepsilon \in (0,t)$ such that $p(s,X) \ge p(t,X)/2$ for $t - \varepsilon \le s \le t$. Thus

$$v(t,X) \ge \int_{t-\varepsilon}^t p(s,X)\, ds \ge \frac{\varepsilon}{2} p(t,X) > 0.$$

This proves the set equality. Together with (3.9) and $\Omega_0 \subset \Omega_t$, it gives

$$\Delta v(t) = \mathbf{1}_{\Omega_t \setminus \Omega_0} = \mathbf{1}_{\{v(t)>0\} \cap (\mathcal{M} \setminus \Omega_0)},$$

which proves (3.8).

Integrating (3.4) in time gives

$$v_0(t, z + C) \le v(t,x,z) \le v_0(t, z - C) \quad \text{for every } (x,z) \in \mathcal{M}.$$

The statement on $\mathbb{R}^2$ follows by periodic lifting. □

4 Eventual regularity and Lipschitz decay

Proof of Main Theorem. The identity at $t = 0$ follows from the definition of Φ. For $t > 0$, Proposition 3.3 and the uniqueness in Proposition 2.2(1) give

$$\Phi(t, \mathcal{O}_0) = \{v(t) > 0\} - te_2 = \Omega_t - te_2 = \mathcal{O}_t.$$

This proves (1.5). Now let $t > \operatorname{osc} h_0$. Observe that

$$F(t,x) - F(0,x) = t + h(t,x) - h_0(x) \geq t - \operatorname{osc} h_0 > 0.$$

Hence [1, Theorem 7.2] implies that $F(t)$, and therefore $h(t)$, is real analytic whenever $t > \operatorname{osc} h_0$. The quantitative estimate in the proof of that theorem also gives, for any fixed $T > \operatorname{osc} h_0$,

$$\sup_{t \geq T} \|\partial_x^2 h(t)\|_{L^\infty(\mathbb{T})} \leq \frac{C(L)}{T - \operatorname{osc} h_0}, \qquad L = \|\partial_x h_0\|_{L^\infty(\mathbb{T})}.$$

Hence, $\sup_{t \geq T} \|\partial_x h(t)\|_{C^{0,1}} < \infty$. In particular, for every $\beta \in (0,1)$,

$$\sup_{t \geq T} \|h(t)\|_{C^{1,\beta}} < \infty.$$

Finally, [3, Proposition 3.2] shows that, for any $\alpha \in (0,1)$,

$$\left\| h(t) - \fint_{\mathbb{T}} h_0(x)\, dx \right\|_{C^\alpha}$$

decays exponentially as $t \to \infty$. Interpolating this decay with the uniform $C^{1,\beta}$ bound above, we conclude that

$$\|\partial_x h(t)\|_{L^\infty}$$

also decays exponentially as $t \to \infty$. □

A The Dirichlet–Neumann operator

B Continuity of the extended pressure

Proof. After discussing the results of this note with my advisor, Jiajun Tong, I came to view them as largely expected. I therefore omit the two remaining appendices, as the corresponding arguments are standard. □

References

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